

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

Xingqiang Chen and Qixing Wang

Department of Physics

Xiamen University

Xiamen 361005, China

(Dated: July 11, 2026)

Substitutional doping and biaxial strain are jointly used to tune covalent molecular networks, yet total energies are still often approximated by superposing results from separate composition-only and strain-only density-functional theory calculations. We study quasi-hexagonal C₆₀ graphullerene with the PBE+D3 functional and define a synergy order parameter $\mathcal{S}$ (meV/atom) for the total-energy cross term missing from sequential strain–doping scans. On B/N/P tetramers and periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$) under matched PBE+D3, linear strain coefficients α and $|\mathcal{S}|$ decouple on rigid tetramer grids (Table I), while phosphorus exhibits the largest joint-load cross term on periodic cells, with $|\mathcal{S}|$ reaching 31.9 meV/atom at $n=1$ under +3% biaxial strain. These rigid-strain values represent upper bounds on equilibrium coupling; ionic relaxation can substantially reduce or even reverse the synergy magnitude. Our results demonstrate that stability ranking under load requires explicit coupled strain–doping evaluations for size-mismatched dopants and large biaxial strain, rather than additive composition–strain models used in sequential high-throughput screening.

I. INTRODUCTION

Covalent molecular networks—from graphullerenes to covalent organic frameworks—are often tuned jointly by substitutional chemistry and mechanical load, yet many high-throughput pipelines still construct $E_{\text{tot}}(\epsilon, \delta)$ by combining *separate* doping-only and strain-only density-functional theory runs [1–3]. Defect formation energies and substitutional stabilities under strain are classical questions in defect physics, but most workflows still assume that composition-only and strain-only increments superpose linearly—an assumption that breaks down when dopant-induced internal stress couples nonlinearly to external load. The omitted cross term is a systematic bias in sequential screening: it arises whenever a dopant introduces a *local chemical-mismatch stress field* that couples nonlinearly to an external biaxial load, so additive composition–strain models can incorrectly rank relative stability even when dopant ordering at $\epsilon=0$ is unchanged. Quasi-hexagonal (qHP) C₆₀ graphullerene [4, 5] is a covalently linked molecular network whose elastic, optical, and transport properties have been surveyed extensively [5–8], yet prior work does not report a periodic total-energy cross term under joint biaxial (ϵ, δ) load. Ab initio models of pristine qHP networks report intrinsic intramolecular strain from the eight-bond hexagonal connectivity [9] ($\sim 0.1\text{--}0.2$ eV-scale cage distortions relative to isolated C₆₀ in prior surveys), so external biaxial load couples to an intrinsically pre-strained, *soft* molecular cage. Eshelby-type linear superposition assumes a linear elastic continuum; because the pre-strained fullerene cage can enter a nonlinear elastic regime, additive superposition of defect and external fields is not guaranteed. Beyond pristine monolayers, strain and substitution independently reshape band edges [2, 10, 11], and substitutional B/N engineering can yield spin-polarized half-semiconductor gaps [12]. Related 2D fullerene networks show anisotropic mechanical and tribological re-

sponse under load [13, 14], and endohedral qHP C₆₀ cage filling reshapes band gaps [15]. None of these reports a periodic total-energy cross term under joint biaxial (ϵ, δ) load. Non-additive coupling is documented in other two-dimensional networks [1, 16]; a quantitative *total-energy* synergy order parameter $\mathcal{S}$ [Eq. (1)] remains unquantified for substitutional B/N/P qHP C₆₀ tetramers and $n = 1\text{--}8$ supercells.

We address three questions on this *class* of covalent molecular networks (qHP C₆₀ as the validation platform): (1) how donor vs. acceptor dopants produce distinct mechanical strain responses and π -electron filling under the same PBE+D3 protocol; (2) how large the omitted non-additive cross term $|\mathcal{S}|$ is under joint load; and (3) whether $|\mathcal{S}|$ scales with dopant–host size mismatch or originates from π -electron nonlinearities. We define $\mathcal{S}$ on periodic PBE+D3 grids at +3% biaxial strain and compare matched-functional tetramer α and $\mathcal{S}_{+3\%}^{\text{tet}}$ on an independent placement map (Table I) to periodic $\mathcal{S}(n)$. The largest periodic $|\mathcal{S}|$ at fixed coordinates (up to 31.9 meV/atom for P at $n=1$) does not track tetramer $|\alpha|$ monotonically—*decoupling* means that linear α cannot quantitatively predict cross-term magnitude or sign. Local dopant distortion and global biaxial strain therefore interact nonlinearly in total energy, so relative stability margins from $\epsilon = 0$ substitution energies or strain-only increments may not hold under coupled loading.

II. COMPUTATIONAL METHODS

A. Electronic structure method

All total-energy calculations use CP2K [17] with PBE [18], Grimme DFT-D3 (Becke–Johnson damping) [19], Goedecker–Teter–Hutter (GTH) pseudopotentials, and the DZVP-MOLOPT-SR-GTH atom-centered basis. We adopt PBE+D3 rather than the Koopmans-

compliant rVV10 protocol of Ref. 5 to enable large-scale (ϵ, δ) grids; PBE underestimates absolute gaps, but $\mathcal{S}$ is a same-functional total-energy difference so systematic gap errors largely cancel. Plane-wave cutoffs are 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$; a single-point 400 Ry control at $6 \times C_{60}$ N and +3% differs by ~ 0.057 meV/atom from the 350 Ry energy at fixed geometry. Unless noted otherwise, the SCF threshold is $10^{-6} E_h$.

B. Strain and doping models

Biaxial strain is defined as $\epsilon = (a - a_0)/a_0 \times 100\%$, where a (a_0) is the strained (pristine equilibrium) in-plane lattice constant; dopant species $\delta \in \{B, N, P\}$. The synergy order parameter is the calligraphic symbol $\mathcal{S}$ [Eq. (1)], in meV per atom. π -projected densities of states are written π -DOS.

Biaxial strain scales the in-plane lattice vectors of the pristine equilibrium cell at fixed fractional coordinates (rigid-strain protocol). Reported α and $\mathcal{S}$ under this protocol are upper bounds on equilibrium magnitudes, because ionic relaxation can lower the energy through structural reconstruction (Sec. IID).

Tetramer models ($4 \times C_{60}$) substitute B, N, or P at nominal ~ 5 at.% on a fixed pseudorandom carbon-site map; configurations are not thermodynamically optimized. Periodic supercells place one substituent per C_{60} (~ 1.67 at.%) on a fixed production site map shared across $n = 1-8$. Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain) is obtained from rigid PBE+D3 tetramer grids; positive α means tensile strain increases the total energy. The substitution energy difference $\Delta E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is not a thermodynamic formation enthalpy (no elemental chemical potentials).

C. Synergy order parameter

We quantify strain-doping cross terms with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta) = [E(\epsilon, \delta) - E(0, \text{pristine})]/N_{\text{atoms}}$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times C_{60}$. The bracketed term is the additive prediction from separate strain-only and doping-only scans; $\mathcal{S} = 0$ under perfect additivity, $\mathcal{S} < 0$ indicates cooperative stabilization, and $\mathcal{S} > 0$ indicates that the coupled corner is less stable than the additive prediction. Periodic $\mathcal{S}$ values are taken from the +3% size-scaling grid (Table IV).

D. Validation benchmarks

Three benchmarks govern interpretation of size-scaling trends: alternate dopant placement (Table I), ionic relaxation of the tetramer P four-corner grid (Table II), and the single-point cutoff control described above.

1. Alternate-placement benchmark

An independent dopant configuration at the same nominal ~ 5 at.% loading repeats the B/N/P strain grid under rigid PBE+D3 (Table I). Main-text quantitative α and $\mathcal{S}_{+3\%}^{\text{tet}}$ use this alternate column together with the reference-placement map in Fig. 1(b). At periodic $n=4$, two independent site maps repeat the four-corner Eq. (1) audit; the $|\mathcal{S}|$ ranking at +3% matches the production map in Table IV.

2. Ionic-relaxation benchmark

Fixed-cell BFGS geometry optimization is performed on the tetramer pristine/ P corners at $\epsilon=0$ and +3% (force thresholds 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr; displacement thresholds 3.0×10^{-3} and 1.5×10^{-3} bohr). A matched-functional rigid PBE+D3 control at the relaxed coordinates quantifies how $\mathcal{S}$ changes upon relaxation (Table II). The same BFGS thresholds are used for the periodic $n=1$ P four-corner relaxation (Sec. III C 1).

3. Hirshfeld population along strain

Hirshfeld atomic charges are computed along $\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$ on periodic $n=1$ B/N/P cells (spin-polarized KS for odd electron counts). They complement geometric $\bar{d}$ in Table III.

III. RESULTS

We first present electronic structure and linear strain response, then focus on the non-additive synergy order parameter $\mathcal{S}$.

A. Electronic structure under strain and doping

Figure 1(a) summarizes tetramer HOMO–LUMO gaps (archived tetramer electronic-structure set). At $\epsilon=0$, B doping results in a nearly closed π gap ($E_g \approx 0.03$ eV), N doping pins the Fermi level inside the conduction band ($E_g \approx -0.14$ eV), and P remains gapped ($E_g \approx 0.05$ eV); tensile strain narrows the pristine gap (~ 23 meV/% on the three available strain points in the dataset). Donor

TABLE I. Tetramer rigid-strain placement sensitivity (PBE+D3). Fig. 1(b) compares reference-placement α (+58.7, -297.5, +456.0 meV/% for B, N, P) with the alternate column. $\mathcal{S}_{+3\%}^{\text{tet}}$ follows Eq. (1) on the rigid tetramer four-corner grid ($N=240$). Periodic $\mathcal{S}(n)$ is in Table IV.

Dopant	Placement	α (meV/%)	$\mathcal{S}_{+3\%}^{\text{tet}}$ (meV/atom)
B	alternate	+21.4	-2.6
N	alternate	+45.4	+12.6
P	alternate	+989.6	+41.4

TABLE II. Ionic-relaxation benchmark for the tetramer P -substitution synergy grid (four fixed-cell geometry-optimization corners: pristine and P at $\epsilon=0$ and +3%, same biaxial cell). Relaxed corners and the matched-functional ratio use PBE+D3 ($\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ meV/atom; $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.9$; sign not preserved). $\mathcal{S}$ follows Eq. (1) with $N=240$ atoms.

Quantity	Rigid (PBE+D3 matched)	Relaxed (geometry opt.)
$E(\text{pristine}, \epsilon=0) (E_h)$	—	-1363.9551
$E(\text{pristine}, \epsilon=3\%) (E_h)$	—	-1363.9550
$E(P, \epsilon=0) (E_h)$	—	-1374.4415
$E(P, \epsilon=3\%) (E_h)$	—	-1374.4615
$\mathcal{S}$ (meV/atom)	+1.19	-2.28
Sign of $\mathcal{S}$ vs. rigid	+	- (not preserved)

vs. acceptor ordering of the $\epsilon=0$ gaps tracks distinct mechanical response in Sec. IIIB. The four-panel π -DOS stack (Fig. 2) shows N-induced states crossing the Fermi level at $\epsilon=0$ (metallic pinning), a narrow π gap for B ($E_g \approx 0.03$ eV), and a wider gap for P—consistent with excess vs. deficient π -electron count at the dopant cage.

B. Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/% on the verified PBE+D3 grids). Rigid PBE+D3 α on the reference placement map gives +58.7, -297.5, and +456.0 meV/% for B, N, and P [Fig. 1(b)]; these share the dopant-site map with the periodic $\mathcal{S}$ audit. On the independent alternate placement (Table I), the same protocol gives +21.4, +45.4, and +989.6 meV/%— α exhibits strong dopant-placement dependence while $\mathcal{S}_{+3\%}^{\text{tet}}$ remains on the order of 10 meV/atom for all three dopants. The largest periodic $|\mathcal{S}|$ at $n=4$ and +3% strain is 23.7 meV/atom for P (Table IV), while $|\alpha_P|$ on the alternate tetramer map is placement-sensitive—consistent with α - $|\mathcal{S}|$ decoupling. Ranking dopant stability solely by zero-strain substitution energies can be misleading under applied strain for close competitors. Table III links mechanical spread to *tetramer* local geometry ($\sim 5\%$ nominal doping, multiple sites): P shows negligible strain-induced change in the mean nearest-neighbor C-C distance ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å vs. $\sim 8.5 \times 10^{-3}$ Å for B at +3% load), indicating a negligible uniform stretching of nearest-neighbor bonds rather than absence of cage buckling. Complementary Hirshfeld charges on the periodic $n=1$ B/N/P strain paths probe charge redistribution along the same load path.

C. Non-additive strain-doping coupling

Figure 1(c) tracks protocol $\mathcal{S}(n)$ at +3% biaxial strain. At $n = 4$ under the fixed periodic production map (Table IV; two independent $n=4$ placement maps preserve $P > B > N$), $\mathcal{S} = -6.2$ meV/atom (B), +3.0 meV/atom (N), and -23.7 meV/atom (P); P shows the largest magnitude despite α - $|\mathcal{S}|$ decoupling on the verified PBE+D3 tetramer map (Table I). On the alternate tetramer map (Table I), P still carries the largest $\mathcal{S}_{+3\%}^{\text{tet}}$ (+41.4 meV/atom) while $|\alpha|$ is placement-sensitive (+989.6 meV/%); decoupling means linear α cannot predict cross-term magnitude or sign. For $n \leq 4$ at matched 400 Ry cutoffs, $|\mathcal{S}|$ reaches 31.9 meV/atom for P at $n = 1$ and remains on the order of 10 meV/atom for B and P at $n = 2-4$ (Table IV). Within $n \leq 4$, $|\mathcal{S}|$ is not monotonic in n (e.g., B: $8.4 \rightarrow 2.2 \rightarrow 6.2$ meV/atom at $n=1, 2, 4$), reflecting finite-cell dopant-dipole interactions and periodic image effects rather than an asymptotic size limit; we report protocol values at each supercell without extrapolating $\mathcal{S}_\infty$. Rows with $n \geq 6$ (350 Ry) appear in Supplemental Material Table S1. Negative $\mathcal{S}$ (B, P at $n = 4$) means coupled tension *stabilizes* the doped cell relative to the additive estimate; positive $\mathcal{S}$ (N at $n = 4$) means sequential screens *overestimate* stability under load.

1. Ionic-relaxation checkpoint

Table II benchmarks four fixed-cell geometry-optimization corners on the tetramer P -substitution grid (pristine and P at $\epsilon=0$ and +3%). For the pristine tetramer, relaxed total energies at $\epsilon=0$ and +3% are -1363.9551 and -1363.9550 E_h ($\Delta E \approx 1.2 \times 10^{-4} E_h$).

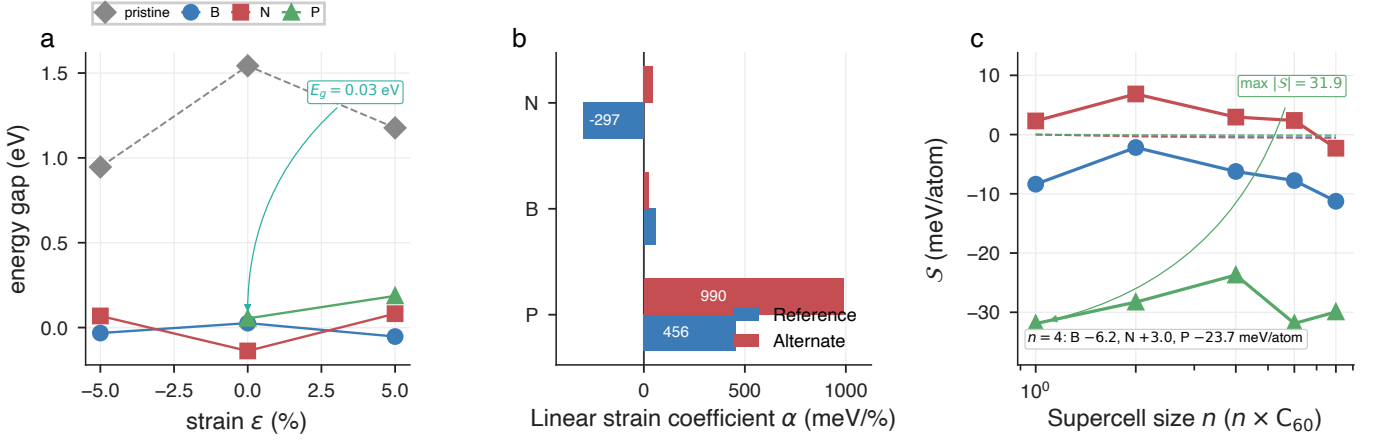


FIG. 1. Coupled (ϵ, δ) response in doped qHP C_{60} . (a) HOMO-LUMO gap vs. strain (tetramer electronic archive; PBE). (b) Mechanical α on reference (blue) vs. alternate (red) tetramer placement maps (PBE+D3 rigid; Table I). (c) Periodic $S(n)$ at +3% biaxial strain (PBE+D3 rigid; $n \leq 4$ core grid; peak $|S|$ at P, $n=1$).

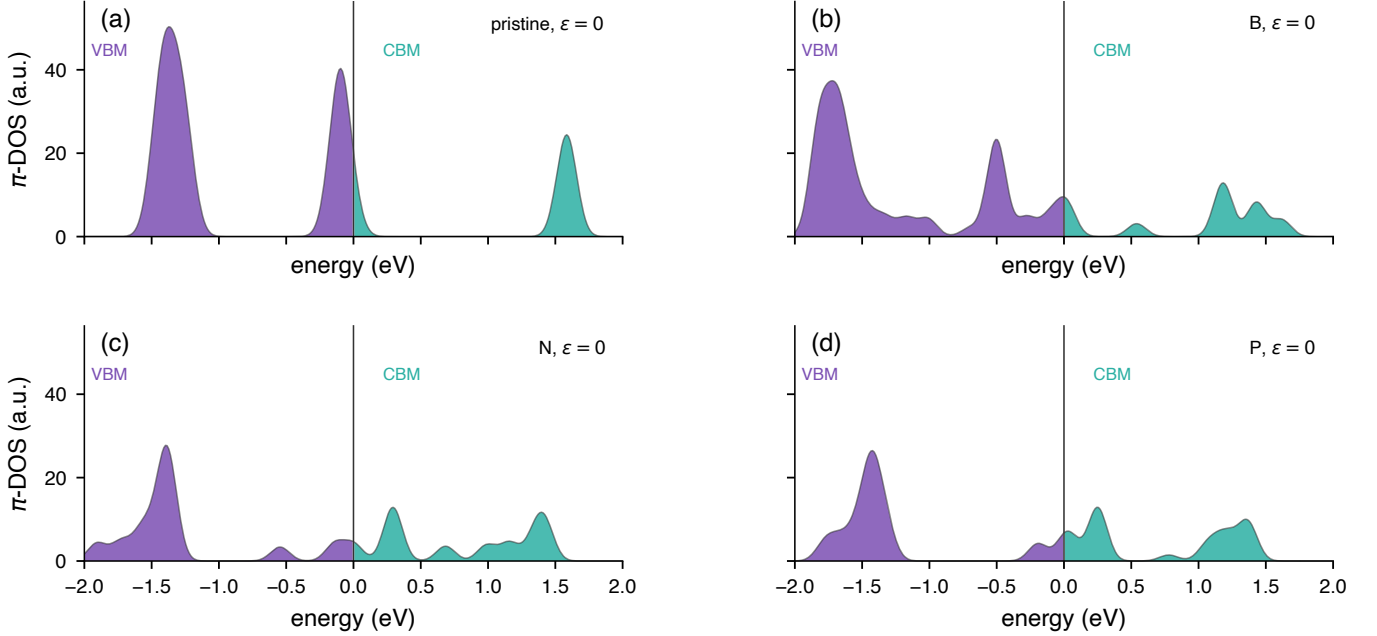


FIG. 2. π -projected density of states (π -DOS) for pristine and B/N/P-doped tetramers at $\epsilon=0$ (PBE+D3). Panels (a)–(d) share a common vertical scale; energy is relative to the Fermi level at 0 eV (Capobianco color scheme [5]).

The P -doped corners give $E(P, \epsilon=0) = -1374.4415$ and $E(P, \epsilon=3\%) = -1374.4615 E_h$. Matched-functional PBE+D3 rigid single-points at relaxed coordinates yield $\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ meV/atom versus $\mathcal{S}_{\text{relaxed}} = -2.28$ meV/atom (sign not preserved; $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.9$; Table II). Matched-functional fixed-cell geometry optimization on the periodic $n=1$ P grid (PBE+D3; four corners) gives $\mathcal{S}_{\text{rigid}} = -31.9$ meV/atom versus $\mathcal{S}_{\text{relaxed}} \approx +0.1$ μ eV/atom ($|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}| \lesssim 10^{-5}$), so ionic relaxation nearly eliminates the rigid periodic cross term in this cell.

IV. DISCUSSION

A. Literature context

Relative to strain-only, doping-only, mobility, or mechanical studies in qHP and related fullerene networks [1, 2, 5–7, 9–15, 20–22], the distinction here is a periodic *total-energy* audit of joint biaxial (ϵ, δ) load [Eq. (1)] (Table IV). Foundational DFT surveys of qHP C_{60} monolayers emphasize anisotropic elastic moduli and optical gaps [6] rather than matched four-corner total-energy grids under joint substitutional and mechanical load.

TABLE III. Local dopant–cage structure from archived *tetramer* XYZ ($\sim 5\%$ nominal doping; qualitative proxy for mismatch trends). Mean nearest-neighbor C distance $\bar{d}$ in Å; standard deviation $\sigma(\bar{d})$ across dopant sites; covalent-radius excess vs. C in pm. Strain-induced change $\Delta\bar{d} = \bar{d}(\epsilon = +3\%) - \bar{d}(\epsilon = 0)$ tracks the uniform nearest-neighbor stretch component.

Dopant	$\bar{d}(\epsilon=0)$ (Å)	$\bar{d}(\epsilon = +3\%)$ (Å)	$\Delta\bar{d}$ (Å)	$\sigma(\bar{d})$ @ +3% (Å)	Δr_{cov} (pm)
B	1.384	1.393	+0.0085	0.019	+7
N	1.393	1.402	+0.0087	0.024	−6
P	1.384	1.385	+0.00003	0.0004	+30

TABLE IV. Periodic synergy audit at +3% biaxial strain (PBE+D3 rigid protocol, reference placement; core $n \leq 4$ at 400 Ry). $\mathcal{S}$ in meV/atom from Eq. (1). Rows with $n \geq 6$ (350 Ry) are in Supplemental Material Table S1.

n (C ₆₀ units)	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
1	B	+3	−8.38
1	N	+3	+2.30
1	P	+3	−31.91
2	B	+3	−2.18
2	N	+3	+6.85
2	P	+3	−28.29
4	B	+3	−6.23
4	N	+3	+2.95
4	P	+3	−23.71

Strain-engineering roadmaps for generic two-dimensional materials [3] likewise foreground band-edge and mobility levers under separate mechanical or chemical protocols. Graphendofullerene derivatives report Curie temperatures tunable through *either* biaxial strain or electrostatic doping applied in isolation [11], reinforcing that joint (ϵ, δ) total-energy corners are not implicit in sequential protocol designs. In other 2D hosts, substitutional defects can raise or lower formation energies under biaxial strain depending on dopant–host size mismatch [22]; spin-resolved impurity engineering on qHP graphullerene instead targets half-semiconductor gaps [12]; we instead quantify the omitted additive cross term $\mathcal{S}$ in total energy.

B. Origins of α – $\mathcal{S}$ decoupling

Substitutional B, N, or P can be viewed as embedding a *localized internal strain field* ϵ_{int} into an already distorted qHP cage baseline [9]. Table III quantifies the chemical-mismatch scale by covalent-radius excess Δr_{cov} (+7, −6, and +30 pm for B, N, and P vs. C) and by mean dopant–C distance $\bar{d}$ at $\epsilon=0$. External biaxial strain ϵ_{ext} is a uniform macroscopic load on the periodic cell. Sequential high-throughput models implicitly assume that responses to ϵ_{int} and ϵ_{ext} superpose linearly; $\mathcal{S}$ is the total-energy evidence that this superposition fails when both fields are present. In the Eshelby picture, a small substitutional defect in a linear elastic continuum produces a strain field that superposes with external load; qHP C₆₀ falls outside this linear elastic regime because the cage is *soft* and pre-strained [9], so second-order defect–elastic work can exceed the increment predicted by linear $\alpha \times \epsilon$ scaling

alone.

N introduces excess π electrons that strengthen bonding under tensile strain (negative α), whereas B creates electron deficiency and antibonding π^* character that rises with lattice expansion (positive α). Quantitatively, the $\epsilon=0$ gap ordering ($E_g^{\text{N}} < E_g^{\text{B}} < E_g^{\text{P}}$ within PBE; Sec. III A) and the dopant-resolved π -DOS stack in Fig. 2 (N states at E_F ; B near-gap; P gapped) align with donor vs. acceptor π -electron counting, while total-energy α spans ~ 970 meV/% between B and P on the verified PBE+D3 alternate tetramer map (Table I). α – $\mathcal{S}$ *decoupling* does not assert anticorrelation between $|\alpha|$ and $|\mathcal{S}|$: on the alternate PBE+D3 map (Table I), P has both the largest $|\alpha|$ and the largest $\mathcal{S}_{+3\%}^{\text{tet}}$. The claim is that linear strain slopes from a one-parameter scan cannot quantitatively predict four-corner cross-term magnitude or sign because α is placement-sensitive at ~ 5 at.% loading whereas $|\mathcal{S}|$ remains on the order of 10 meV/atom on matched-functional tetramer and periodic audits.

The tetramer linear coefficient α and the periodic cross term $\mathcal{S}$ probe different energy partitions: α is the slope of total-cell energy under uniform lattice expansion at $\sim 5\%$ multi-site doping, whereas $\mathcal{S}$ is the omitted four-corner cross term per atom at ~ 1.67 at.% under periodic boundary conditions. They therefore need not share sign even for the same element—a negative α (tension lowers total energy in a strain-only scan) can coexist with positive $\mathcal{S}$ (the coupled corner is less stable than an additive model predicts), as for N at $n=4$. Any N sign flip between $n=6$ and $n=8$ lies outside the matched-cutoff $n \leq 4$ core grid and is excluded from design rules.

C. Geometric vs. electronic coupling mechanisms

For *size-mismatched* P, the dopant introduces compressive local misfit in the cage ($\Delta r_{\text{cov}} = +30$ pm) while tensile ϵ_{ext} partially relieves that misfit without proportionally stretching nearest-neighbor bonds ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å at +3%; Table III). The resulting negative $\mathcal{S}$ (cooperative stabilization relative to the additive estimate) is consistent with nonlinear defect–elastic coupling: external strain couples to *pre-existing cage buckling* rather than to uniform bond stretching. P carries the largest covalent-radius excess and pre-existing cage distortion at $\epsilon=0$, yet shows the smallest strain-induced change in mean nearest C distance and the smallest site-to-site spread at +3% ($\sigma(\bar{d}) \approx 4 \times 10^{-4}$ Å vs. $\sim 2 \times 10^{-2}$ Å for B/N; Table III), yielding the largest $|\mathcal{S}|$ at $n=4$ (23.7 meV/atom; Table IV). For B and N, $|\Delta r_{\text{cov}}| \lesssim 10$ pm and $\Delta \bar{d}$ tracks ϵ_{ext} more uniformly ($\sim 8.5 \times 10^{-3}$ Å at +3%), so $\mathcal{S}$ is set primarily by electronic nonlinearity—donor vs. acceptor π -electron counting and Fermi-level pinning—rather than by large-amplitude cage reshaping. At $n=4$ the coupled correction for P exceeds the rigid per-atom strain increment from tetramer α by about two orders of magnitude ($|\mathcal{S}| \gg \alpha \times 3\%/N_{\text{atoms}}$), even though $|\alpha|$ ranks P as the weakest strain responder on the tetramer grid.

For the matched-functional tetramer *P* rigid grid, Grimme D3 dispersion energies at the four corners give $\mathcal{S}_{\text{vdW}} \approx +0.33$ meV/atom—orders of magnitude below $|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.2$ meV/atom (Table II)—so vdW non-additivity does not drive the reported cross terms. On the periodic $n=1$ P grid, Hirshfeld charge on P decreases from $+0.063e$ at $\epsilon=0$ to $+0.056e$ at +3% while tetramer $\Delta \bar{d}$ for P remains below 10^{-4} Å at +3% load (Table III). B and N Hirshfeld charges shift from $+0.119e$ and $+0.069e$ at $\epsilon=0$ to $+0.111e$ and $+0.057e$ at +3% ($\sim 7\%$ and $\sim 18\%$ fractional $\Delta q/q$, vs. $\sim 11\%$ for P). Thus N shows the largest fractional charge response yet the smallest $|\mathcal{S}|$ at $n=4$, whereas P combines a moderate Δq with vanishing $\Delta \bar{d}$ and the largest cross term—charge redistribution alone does not rank $|\mathcal{S}|$; geometric pre-strain remains necessary for the P channel.

Together, these observations suggest a taxonomy for covalent molecular networks: *electronic* donor/acceptor control of α ; *structural* local-distortion coupling to external strain (P); and *finite-size* electrostatic dilution (N at large n).

D. General design rules for covalent networks

Using $|\Delta r_{\text{cov}}|$ from Table III as a single mismatch coordinate, periodic $|\mathcal{S}|$ spans 2.3–8.4 meV/atom for B and N ($|\Delta r_{\text{cov}}| \lesssim 10$ pm) but reaches 31.9 meV/atom for P ($|\Delta r_{\text{cov}}|=30$ pm; Table IV). N breaks a naive “ $|\mathcal{S}| \propto (\Delta r)^2$ ” law at comparable mismatch to B because its cross term is electronic (positive $\mathcal{S}$ at $n=4$) rather than geometric. The data support a *two-regime*

design rule: $|\Delta r_{\text{cov}}| \gtrsim 20$ pm flags the geometric-nonlinear regime where coupled audits are mandatory (P); $|\Delta r_{\text{cov}}| \lesssim 15$ pm leaves $|\mathcal{S}|$ at few-meV/atom under +3% load but still permits sign conflicts between α and $\mathcal{S}$ (B, N). These thresholds are calibrated on qHP C₆₀; rigid graphene-like hosts may require higher mismatch before geometric nonlinearity dominates.

At $|\Delta r_{\text{cov}}| \lesssim 10$ pm and +3% biaxial load, $|\mathcal{S}| \lesssim 10$ meV/atom on the $n=1$ audit grid; sequential pipelines may be adequate for coarse ranking but not for close E_{sub} competitors. Pristine tetramers show $|\alpha| \approx 1$ meV/%, so sub-1% loads keep linear strain increments small; at +3% and above, coupled corners for size-mismatched dopants are advisable. qHP C₆₀ combines interfullerene covalent links with soft intramolecular cages and intrinsic intramolecular strain [9], amplifying local-distortion coupling relative to rigid 2D lattices; extrapolation to graphene-like hosts requires separate benchmarks. Tetramer α and $\mathcal{S}_{+3\%}^{\text{tet}}$ are not interchangeable with periodic $\mathcal{S}(n)$ because concentration ($\sim 5\%$ vs. ~ 1.67 at.%) and boundary conditions differ; non-additivity is assessed only through Eq. (1) on matched four-corner grids.

At $n=4$ and +3% strain the coupled correction is $|\mathcal{S}| = 23.7$ meV/atom for P vs. 6.2 meV/atom for B—sequential additive estimates differ by approximately a factor of four in the load-dependent margin under the same PBE+D3 periodic protocol. Where $\mathcal{S} < 0$ (B and P at $n=4$), coupled tension can stabilize doped cells beyond additive predictions; where $\mathcal{S} > 0$ (N at $n=4$), sequential screens overestimate stability under load. For B at $n=4$, negative $\mathcal{S}$ coexists with positive α ; for N at $n=4$, negative tetramer α and positive $\mathcal{S}$ present a second sign conflict. Gap closing [Fig. 1(a)] does not track $|\mathcal{S}|$, so photonic or transport targets cannot be inferred from total-energy $\mathcal{S}$ alone.

E. Experimental signatures

Three classes of laboratory observables could test the stress-field picture without resolving total energies directly. **Raman or IR cage modes:** under joint doping and strain, mode shifts should deviate from the sum of doping-only and strain-only shifts when $|\mathcal{S}|$ is large—especially for P, where tensile load couples to pre-existing cage distortion rather than uniform bond stretching (Table III). **Scanning tunneling spectroscopy:** dopant-site local density of states should evolve nonlinearly with strain when Hirshfeld charge redistributes without proportional $\Delta \bar{d}$ (Sec. IV C); N pinning near E_F vs. gapped B/P provides contrasting signatures. **Thermal or chemical stability:** for $\mathcal{S} < 0$ (B, P at $n=4$), tension should enhance apparent doping stability relative to additive thermochemical estimates; for $\mathcal{S} > 0$ (N at $n=4$), the opposite bias is expected. Quantitative links require matched sample chemistry and strain calibration; we quote DFT protocol bounds only.

F. Limitations and outlook

Main-text α and $\mathcal{S}$ use PBE+D3 under the rigid-strain protocol (Sec. II B); ionic relaxation can reverse or nearly extinguish $\mathcal{S}$ (Table II; periodic $n=1$ P). Periodic $\mathcal{S}(n)$ magnitudes are configuration-specific on a fixed production site map (two independent $n=4$ maps preserve $P>B>N$; Table IV). Core claims use matched 400 Ry for $n \leq 4$; $n \geq 6$ production points use 350 Ry. Table III geometry is a tetramer proxy; the geometric vs. electronic channel assignment rests on $\Delta\bar{d}$, Δr_{cov} , and Hirshfeld Δq rather than a full energy decomposition. Because $\mathcal{S}$ is a same-functional difference, absolute errors in substitution energy do not affect our conclusion of non-additive coupling. Future work includes Mayer/Bader strain-path partitioning and placement averaging on periodic cells.

V. CONCLUSIONS

We quantify how substitutional chemistry and biaxial strain couple nonlinearly in covalent molecular networks: the omitted total-energy cross term $\mathcal{S}$ arises when a dopant-induced internal stress field meets an external load, not from a graphullerene-specific numerical artifact. On qHP C_{60} as the validation platform, donor and acceptor dopants produce distinct π -electron filling and gap response, with placement-sensitive PBE+D3 α on tetramer maps (Table I), while omitted cross terms reach the meV/atom scale on the periodic grid ($|\mathcal{S}|$ up to 31.9 meV/atom for P at $n=1$ at fixed coordinates;

Table IV). For P, $|\mathcal{S}|$ tracks local cage distortion and mismatch (Table III) rather than tetramer $|\alpha|$. Two mechanisms emerge—electronic donor/acceptor π control (B, N) and geometric local-distortion coupling for size-mismatched P—with a practical mismatch threshold ($|\Delta r_{\text{cov}}| \gtrsim 20$ pm) separating few-meV from tens-of-meV $|\mathcal{S}|$ under +3% load. Coupled (ϵ, δ) screening is advisable for size-mismatched substituents, +3% and larger loads, and close substitution-energy competitors; Raman, STS, and stability trends in Sec. IV E offer experimental tests of the stress-field picture.

ACKNOWLEDGMENTS

We acknowledge computational resources and support from Xiamen University.

DATA AVAILABILITY

The data that support the findings of this study are openly available at <https://github.com/chenxingqiang/graphullerene-strain-engineering>, including computational inputs and outputs, tabulated synergy and polaron analyses, and materials to reproduce all figures and tables in the main text. Ionic-relaxation, alternate-placement, Hirshfeld, and size-scaling archives referenced in Tables I–IV are included in the public release. A tagged release matching the manuscript revision will be issued upon acceptance.

-
- [1] S. Chen, H. Liu, and K. Wang, Untangling the respective effects of heteroatom doping and strain on electronic transport in 2D materials, *Adv. Mater.* **36**, 2301234 (2024).
 - [2] H. Li, M. A. Shakoori, and R. Wang, strain engineering of electronic structure and thermoelectric properties of quasi-hexagonal fullerene monolayer, *J. Appl. Phys.* **136**, 015101 (2024).
 - [3] A. K. Katiyar and J. H. Ahn, strain-engineered 2D materials: challenges, opportunities, and future perspectives, *Small Methods* **9**, 2401404 (2025).
 - [4] Y. Yang, W. Zhang, H. Liu, and K. Li, two-dimensional carbon allotropes with tunable direct band gaps and high carrier mobility, *Appl. Surf. Sci.* **536**, 147815 (2021).
 - [5] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa, electron localization and mobility in monolayer fullerene networks, *Nano Lett.* **24**, 7891 (2024).
 - [6] R. M. Tromer, L. A. R. Junior, and D. S. Galvao, A DFT study of the electronic, optical, and mechanical properties of a recently synthesized monolayer fullerene network, *Chem. Phys. Lett.* **804**, 139925 (2022).
 - [7] R. Lv, J. Xi, M. Chen, and S. Yang, Covalent fullerene networks: preparation, properties and catalytic applications, *Rev. Mater. Res.* **2**, 100195 (2026).
 - [8] P. Silva, R. M. Faria, L. G. Ferreira, and J. L. F. Fatima, large polarons in two-dimensional fullerene networks: the crucial role of anisotropy in charge transport, *Nanoscale* **16**, 2337 (2024).
 - [9] G. Makov and U. Argaman, structure and properties of graphullerene: a semiconducting two-dimensional C_{60} crystal, *npj Comput. Mater.* **9**, 211 (2023).
 - [10] B. Peng and M. Pizzochero, monolayer C_{60} networks: a first-principles perspective, *Chem. Commun.* **61**, 10287 (2025).
 - [11] D. López-Alcalá, Z. Hu, and J. J. Baldoví, graphendo-fullerene: a novel molecular two-dimensional ferromagnet, *Chemical Science* **16**, 7659 (2025).
 - [12] M. A. Khan and M. N. Leuenberger, tuning the electronic and optical properties of impurity-engineered two-dimensional graphullerene half-semiconductors, *Phys. Rev. Mater.* **9**, 034001 (2025).
 - [13] J. Wang, K. Li, and H. Zhang, simulation on stress-related anisotropy of qTP C_{60} and qHP C_{60} : implications for optoelectronic nanodevices, *ACS Applied Nano Materials* **7**, 3456 (2024).
 - [14] F. Qiu, H. Wang, P. Zhang, and Y. Li, atomic-scale insights into the mechanical and tribological properties of graphullerene, *Tribol. Int.* **204**, 110756 (2025).

- [15] M. L. P. Junior, R. M. Tromer, L. A. R. Junior, and D. S. Galvão, endohedral derivatives of the recently synthesized two-dimensional fullerene networks: electronic and optical insights from first-principles calculations, arXiv preprint (2026), arXiv:2603.10142 [cond-mat.mtrl-sci].
- [16] M. Alihosseini and M. Neek-Amal, Strain tuning of optoelectronic properties of covalent organic framework bilayers and heterostructures, *Phys. Rev. B* **108**, 165403 (2023).
- [17] J. VandeVondele, M. Krack, F. Mohamed, M. Parrinello, T. Chassaing, and J. Hutter, Quickstep: Fast and accurate density functional calculations using a mixed gaussian and plane waves approach, *Comput. Phys. Commun.* **167**, 103 (2005).
- [18] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized gradient approximation made simple, *Phys. Rev. Lett.* **77**, 3865 (1996).
- [19] S. Grimme, S. Ehrlich, and L. Goerigk, effect of the damping function in dispersion corrected density functional theory, *J. Comput. Chem.* **32**, 1456 (2011).
- [20] A. Shaikh, J. Wu, and B. Peng, negative and positive anisotropic thermal expansion in 2D fullerene networks, arXiv preprint (2025), arXiv:2504.02037 [cond-mat.mtrl-sci].
- [21] Y.-B. Shi, S.-H. Lv, Z.-F. Shao, H.-K. Dong, S. Cao, and P. Qian, a first-principles study of 1d and 2d C₆₀ nanostructures: strain effects on band alignments and carrier mobility, *J. Phys.: Condens. Matter* **35**, 225701 (2023).
- [22] P. Santra, S. Ghaderzadeh, M. Ghorbani-Asl, H.-P. Komsa, E. Besley, and A. V. Krashenninnikov, strain-modulated defect engineering of two-dimensional materials, *npj 2D Materials and Applications* **8**, 33 (2024).